

WSQ Business Transformation with OpenClaw

Lab Guide

Course Code: TGS-2022015374

Institution: Tertiary Infotech Pte. Ltd.

Instructor: Dr. Alfred Ang

Duration: 1 Day | Total lab time: 280 minutes

Digital Attendance (Mandatory)

It is mandatory to take both AM, PM and Assessment digital attendance for WSQ funded courses.

The trainer will show a digital attendance QR code from the SSG portal. Scan it with your phone.

Ground Rules

- Set your mobile phone to silent mode
- Participate actively – no question is stupid
- Mutual respect: agree to disagree
- One conversation at a time
- Be punctual; return from breaks on time
- 75% attendance is required

Learning Outcomes

SFw: Emerging Technology Synthesis ICT-SNA-3011-1.1

- L01: Deploy OpenClaw on a Hostinger VPS or Docker Desktop local machine
- L02: Connect LLM providers: MiniMax M2.7 (free), OpenAI OAuth, Ollama, Default config
- L03: Set up Telegram (BotFather) and WhatsApp (QR pairing) channels
- L04: Integrate tools: AgentMail, Agent Browser, Firecrawl
- L05: Install and use skills from the ClawHub marketplace
- L06: Schedule tasks with cron jobs and monitor uptime with heartbeat
- L07: Apply 10-step security hardening for OpenClaw
- L08: Access the web dashboard locally and via SSH tunnel
- L09: Optimise costs using model selection and context compaction
- L010: Integrate Google Workspace via service account (Gmail, Calendar, Chat)

About This Lab Guide

This guide accompanies the WSQ Business Transformation with OpenClaw course. Students may choose either lab environment:

- Hostinger VPS – Ubuntu 22.04 LTS, runs 24/7, ideal for production deployment
- Docker Desktop – local machine (Windows 10/11 / macOS 12+ / Ubuntu 22.04+), ideal for learning

Hostinger referral (discount): <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

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Lab 1 – OpenClaw Hosting

Estimated time: 30 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: Option A – Hostinger VPS: `ssh root@YOUR_VPS_IP`
- Step 2: Install: `curl -fsSL https://openclaw.ai/install.sh | bash`
- Step 3: Option B – Docker Desktop (local): `docker pull openclaw/openclaw:latest`
- Step 4: Run container: `docker run -d -p 18789:18789 -v openclaw-data:/root/.openclaw openclaw/openclaw:latest`
- Step 5: Run onboarding: `openclaw onboard` (VPS) or `docker exec -it openclaw openclaw onboard` (Docker)
- Step 6: Verify: `openclaw gateway status` – Gateway: running Port: 18789

Verification

Run all verification checks shown in your lab.md file before moving to the next lab.

Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 2 – OpenClaw Model

Estimated time: 30 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: MiniMax M2.7 (free): openclaw model set minimax --api-key YOUR_KEY --model abab6.5s-chat
- Step 2: OpenAI OAuth step 1: openclaw model set openai --oauth ' copy URL to browser
- Step 3: OpenAI OAuth step 2–4: sign in ' Allow ' paste redirect URL back
- Step 4: Ollama (local): ollama launch openclaw then openclaw model set ollama --host http://localhost:11434
- Step 5: Default config: nano ~/.openclaw/openclaw.json ' set provider + model + apiKey
- Step 6: Test: openclaw model test ' Status: " Connected

Verification

Run all verification checks shown in your lab.md file before moving to the next lab.

Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 3 – OpenClaw Channel

Estimated time: 30 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: Telegram step 1: open Telegram & search @BotFather & send /newbot
- Step 2: Telegram step 2: enter bot name (e.g. Alfred Agent) and username (e.g. alfred_agent_bot)
- Step 3: Telegram step 3: copy bot token 7123456789:AAGxxx!
- Step 4: Register: openclaw channel add telegram --token YOUR_BOT_TOKEN
- Step 5: Start: openclaw channel start telegram
- Step 6: WhatsApp: openclaw channel add whatsapp & scan QR code on phone & openclaw channel start whatsapp
- Step 7: Verify: openclaw channel list & telegram running | whatsapp running

Verification

Run all verification checks shown in your lab.md file before moving to the next lab.

Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 4 – OpenClaw Tools

Estimated time: 30 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: AgentMail: sign up at <https://agentmail.to> – copy API key
- Step 2: Add: openclaw tools add agentmail --api-key YOUR_KEY
- Step 3: Agent Browser: get key at <https://agent-browser.dev>
- Step 4: Add: openclaw tools add agent-browser --api-key YOUR_KEY
- Step 5: Firecrawl: sign up at <https://www.firecrawl.dev> – copy fc-... API key
- Step 6: Add: openclaw tools add firecrawl --api-key YOUR_KEY
- Step 7: Test in chat: 'Check my email' | 'Browse <https://...>' | 'Scrape <https://...>'

Verification

Run all verification checks shown in your lab.md file before moving to the next lab.

Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 5 – OpenClaw Skills

Estimated time: 20 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: Browse marketplace: <https://clawhub.ai>
- Step 2: Install skill: `npx clawhub@latest install web-research`
- Step 3: Docker: `docker exec openclaw npx clawhub@latest install web-research`
- Step 4: Install second skill: `npx clawhub@latest install daily-briefing`
- Step 5: List: `openclaw skills list` – web-research active | daily-briefing active
- Step 6: Test in chat: `/web-research` What is OpenClaw?
- Step 7: Update: `npx clawhub@latest update web-research`

Verification

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Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 6 – Cron Jobs and Heartbeat

Estimated time: 30 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: Create daily cron: `openclaw cron create --schedule '0 9 * * *' --prompt 'Summarise my email' --channel telegram --name morning`
- Step 2: Create weekly cron: `openclaw cron create --schedule '0 8 * * 1' --prompt '/web-research AI news' --channel telegram --name weekly`
- Step 3: List / run now: `openclaw cron list` | `openclaw cron run morning`
- Step 4: Pause / resume: `openclaw cron pause morning` | `openclaw cron resume morning`
- Step 5: Heartbeat: `nano ~/.openclaw/HEARTBEAT.md` ↑ interval: 30m channel: telegram
- Step 6: Or CLI: `openclaw heartbeat enable --interval 30m --channel telegram`
- Step 7: Verify: `openclaw heartbeat status` ↑ Heartbeat: enabled Interval: 30m

Verification

Run all verification checks shown in your lab.md file before moving to the next lab.

Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 7 – OpenClaw Security

Estimated time: 30 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: Step 1 – API keys in env vars: `export MINIMAX_API_KEY='eyJ...' >> ~/.bashrc`
- Step 2: Step 2 – Allowlist: `openclaw channel config telegram --allowlist-add YOUR_TELEGRAM_ID`
- Step 3: Step 3 – Gateway auth: `openclaw config set gateway.auth-token $(openssl rand -hex 32)`
- Step 4: Step 4 – Sandbox exec: `mkdir ~/openclaw-sandbox && openclaw tools config exec --sandbox-dir ~/openclaw-sandbox`
- Step 5: Step 5 – Disable unused tools: `openclaw tools disable exec`
- Step 6: Step 6 – Audit logs: `openclaw logs --last 50 | openclaw logs --export ~/audit.json`
- Step 7: Step 10 – Review: `cat ~/.openclaw/SOUL.md` and `~/.openclaw/AGENTS.md`

Verification

Run all verification checks shown in your lab.md file before moving to the next lab.

Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 8 – OpenClaw Dashboard

Estimated time: 20 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: Local: openclaw dashboard – open `http://localhost:18789` in browser
- Step 2: VPS SSH tunnel: `ssh -L 18789:localhost:18789 root@YOUR_VPS_IP` (keep terminal open)
- Step 3: Then open: `http://localhost:18789` in your LOCAL browser
- Step 4: Explore panels: Overview | Channels | Tools | Skills | Logs | Memory | Cron
- Step 5: Memory panel: view MEMORY.md, SOUL.md, AGENTS.md, HEARTBEAT.md
- Step 6: Gateway restart: Dashboard – Gateway – Restart – wait 5s – refresh

Verification

Run all verification checks shown in your lab.md file before moving to the next lab.

Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 9 – OpenClaw Cost Saving

Estimated time: 20 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: Switch to Haiku (cheapest Claude): `openclaw model set anthropic --model claude-haiku-4-5-20251001`
- Step 2: Claude subscription: `openclaw model set anthropic --subscription` (fixed monthly, unlimited)
- Step 3: Context compaction: `openclaw config set context.compaction enabled`
- Step 4: Compaction threshold: `openclaw config set context.compaction-threshold 50000`
- Step 5: Free testing: `openclaw model set minimax --api-key KEY --model abab6.5s-chat` (no cost)
- Step 6: Zero API cost: `openclaw model set ollama --model openclaw` (runs locally)
- Step 7: Monitor: `openclaw usage stats` | `openclaw usage alert --threshold 5.00 --channel telegram`

Verification

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Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Lab 10 – Google Integration for OpenClaw

Estimated time: 40 minutes

Lab environment: Hostinger VPS (Ubuntu 22.04) OR Docker Desktop (local machine)

Hostinger referral: <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Steps

- Step 1: Step 1 – Google Cloud Console: <https://console.cloud.google.com> – New Project – openclaw-integration
- Step 2: Step 2 – Enable APIs: Gmail API | Google Calendar API | Google Chat API
- Step 3: Step 3 – Service Account: IAM & Admin – Service Accounts – Create – openclaw-agent
- Step 4: Step 4 – JSON key: Keys tab – Add Key – JSON – download file
- Step 5: Step 5 – Upload: `scp file.json root@VPS:/root/.openclaw/gsa.json` (VPS) or `docker cp` (Docker)
- Step 6: Step 6 – Set env: `export GOOGLE_CHAT_SERVICE_ACCOUNT_FILE=/root/.openclaw/gsa.json`
- Step 7: Step 7 – Register: `openclaw tools add google --service-account $GOOGLE_CHAT_SERVICE_ACCOUNT_FILE`
- Step 8: Test: 'Check my Gmail' | 'What meetings do I have tomorrow?'

Verification

Run all verification checks shown in your lab.md file before moving to the next lab.

Troubleshooting

See the Troubleshooting table in your lab.md for common issues and fixes.

Reference Documentation

OpenClaw install: <https://docs.openclaw.ai/install>

Providers: <https://docs.openclaw.ai/providers>

Channels: <https://docs.openclaw.ai/channels>

Tools: <https://docs.openclaw.ai/tools>

Skills: <https://docs.openclaw.ai/skills>

Security: <https://docs.openclaw.ai/gateway/security>

Dashboard: <https://docs.openclaw.ai/dashboard>

ClawHub marketplace: <https://clawhub.ai>

MiniMax (free model): <https://www.minimaxi.com>

Ollama (local model): <https://ollama.com>

AgentMail: <https://agentmail.to>

Firecrawl: <https://www.firecrawl.dev>

Agent Browser: <https://agent-browser.dev>

Google Cloud Console: <https://console.cloud.google.com>

Hostinger VPS (referral): <https://www.hostinger.com?REFERRALCODE=FEGANGCHQ20C>

Final Assessment

Written Assessment (Short Answer Questions) + Oral Questioning

Format: Open Book “ slides, Learner Guide, or approved materials only

Assessment Criteria

- OpenClaw gateway running (VPS or Docker) “ openclaw gateway status
- Agent responds via Telegram and WhatsApp
- Agent uses Firecrawl to summarise a webpage
- Cron job fires and delivers result to channel
- Security settings applied: allowlist and sandbox shown
- Dashboard accessible at <http://localhost:18789>
- Google integration: agent reads Gmail or Calendar event

Support

Email: enquiry@tertiaryinfotech.com

Tel: +65 6100 0613

LMS: <https://ai-lms-tms.tertiaryinfo.tech/>